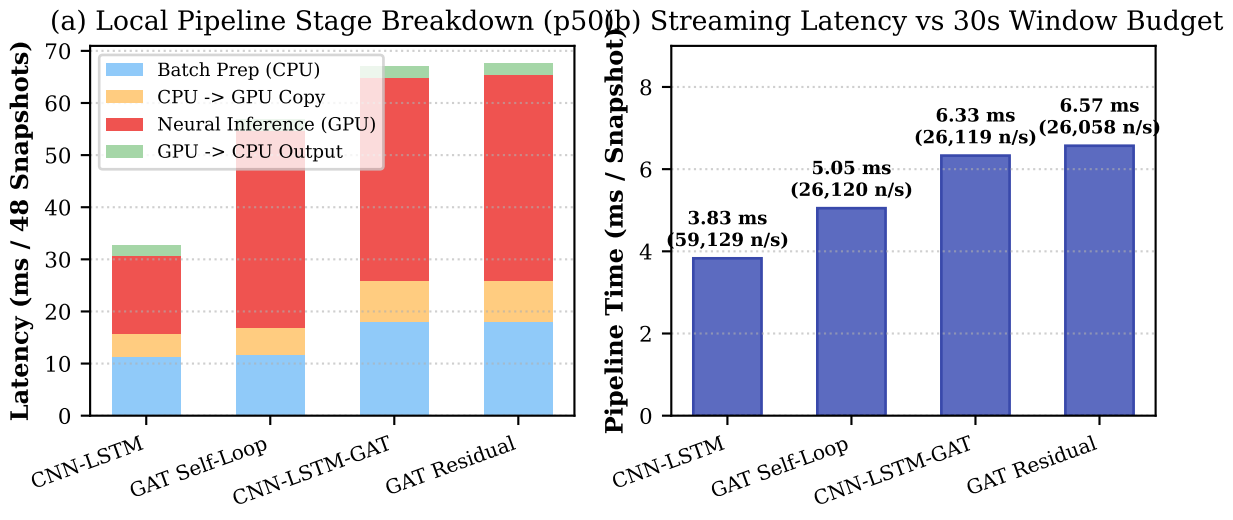
(a) Local Pipeline Stage Breakdown (p50)
Latency (ms / 48 Snapshots)
Batch Prep (CPU)
CPU -> GPU Copy
Neural Inference (GPU)
GPU -> CPU Output
0
10
20
30
40
50
60
70
CNN-LSTM
GAT Self-Loop
CNN-LSTM-GAT
GAT Residual
(b) Streaming Latency vs 30s Window Budget
Pipeline Time (ms / Snapshot)
3.83 ms
(59,129 n/s)
5.05 ms
(26,120 n/s)
6.33 ms
(26,119 n/s)
6.57 ms
(26,058 n/s)
0
2
4
6
8
CNN-LSTM
GAT Self-Loop
CNN-LSTM-GAT
GAT Residual